

MoSAIC: Motion- and Supervision-Aware Inference for Myocardial Scar and Edema Segmentation from Multi-Sequence and Cine CMR

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Abstract. CMR pathology segmentation must account for variable sequences and annotation sets. We present MoSAIC: Motion- and Supervision-Aware Inference for Myocardial Scar and Edema Segmentation. The cascade uses anatomy to condition multi-sequence and cine CMR decoding. Coarse cardiac anatomy supplies a spatial prior for multi-sequence scar decoding, a dedicated edema-zone expert, and a cine scar head using end-diastole (ED)-anchored motion and warped anatomy. Presence-gated encoding and supervision-masked losses represent missing sequences and unannotated classes. The edema expert predicts an injury zone for fusion of scar and edema. On CARE-2026 CARE-Myocardium official validation, the best MoSAIC Dice entries are 0.6965 for MyoPS scar, 0.5983 for MyoPS edema and 0.2069 for CineMyoPS scar. Compared with nnU-Net, these entries improve scar and cine scar Dice and raise three-track mean Dice from 0.492 to 0.501. The retained internal five-fold table requires recovery of its generation provenance; no module-causal or controlled robustness conclusions are drawn.

Keywords: Myocardial pathology segmentation · Multi-sequence CMR · Cine CMR · Motion-derived evidence · Partial supervision.

1 Introduction

After acute myocardial infarction, scar marks irreversibly necrotic myocardium while edema marks salvageable injured tissue. Their relation guides reperfusion assessment [6,9], and CMR resolves them through complementary contrasts: LGE highlights scar [11], T2-weighted imaging highlights edema [1], and bSSFP/C0 supplies anatomy. Automating their joint delineation is the goal of the MyoPS benchmark family [12,28].

The CARE-2026 setting emphasizes realistic heterogeneity in both image evidence and annotation protocols. In multi-sequence CMR, training cases may contain LGE, LGE+C0 or the full LGE+C0+T2 combination; 80 of 220 cases annotate edema, and centers provide different anatomical label sets. Effective training therefore requires the objective to respect the class set that each center annotated. In cine-only CMR, scar is represented primarily through regional

wall-motion abnormality [26,3], making motion a clinically meaningful signal for contrast-free pathology inference.

MoSAIC links coarse anatomy localization to three fine-stage experts (Fig. 1): multi-sequence scar/anatomy decoding, injury-zone prediction (edema $\cup$ scar), and cine scar prediction from ED-anchored motion and warped anatomy. Its contributions are this cascade, presence-gated input handling and supervision masking for heterogeneous labels, and organizer-scored validation. These are architecture and evaluation contributions, without claims of module-specific performance gains.

2 Related Work

MyoPS benchmarks joint scar and edema segmentation using three CMR sequences [28,12]. Related methods address flexible sequence use [18], joint alignment and segmentation [4], and coarse-to-fine localization [25,24]. U-Net and nnU-Net provide generic segmentation backbones [19,10]. Methods for missing modalities represent variable inputs [8,5], while partial-label losses account for unannotated classes [22,27,7]. MoSAIC combines presence gating with per-class supervision masks.

Prior work on contrast-free inference uses cine appearance [26], strain [21,16], or joint motion and segmentation [17]. CineMyoPS combines anatomy with motion [3]; MoSAIC likewise uses motion-derived evidence for its cine scar head.

3 Preliminaries

Problem setup. A dataset is a set of tuples $\mathcal{D} = \{(X_i, Y_i, \mathbf{m}_i, \mathbf{s}_i)\}$. The label map $Y \in \{0, \dots, C\}^{D \times H \times W}$ uses the challenge classes {background, myocardium, LV, RV, edema, scar}. For label fusion, we denote the injury zone as $Z = \text{edema} \cup \text{scar}$ and recover pure edema after scar prediction. Two acquisition regimes occur. In the *multi-sequence* regime $X = (\mathbf{I}_{\text{LGE}}, \mathbf{I}_{\text{C0}}, \mathbf{I}_{\text{T2}})$ and $\mathbf{m} \in \{0, 1\}^3$ records which sequences a center actually acquired, with absent sequences zero-filled. In the *cine* regime $X = (I_0, \dots, I_{T-1})$ is a single bSSFP sequence sampled over the cardiac cycle and anchored at end-diastole (ED). In both regimes $\mathbf{s} \in \{0, 1\}^C$ records which classes the originating center annotated; $s_c = 0$ means the label map is silent about class c , leaving its underlying presence unspecified.

Evaluation. Internal cross-validation reports the Dice similarity coefficient and the 95th-percentile Hausdorff distance (HD95, in mm) computed in the resampled physical space. Organizer-scored validation uses the CARE-Myocardium metrics, Dice and Hausdorff distance (HD, in mm).

4 Method

Stage 1 supplies anatomy and a spatial prior to the multi-sequence scar, edema-zone and cine-motion experts.

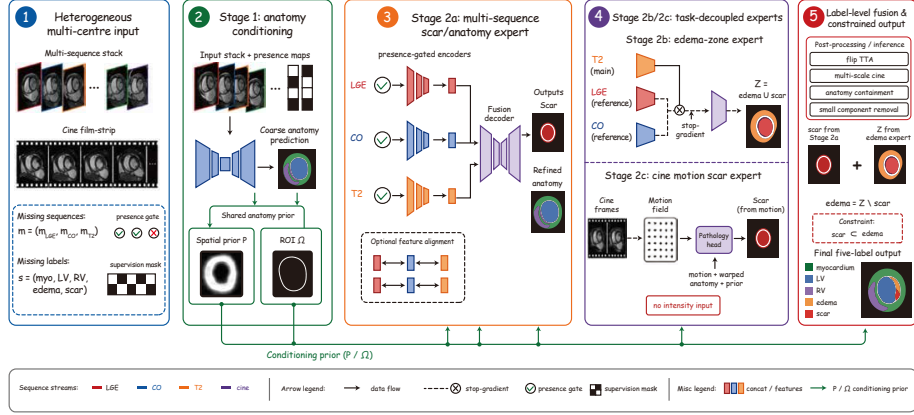


Fig. 1. Overview of MoSAIC. A coarse anatomy stage defines the spatial prior; separate multi-sequence scar, edema-zone, and cine-motion branches produce task-specific predictions that are fused in the challenge label space. Z denotes the injury zone; final pure edema excludes scar.

4.1 Stage 1: Anatomy Conditioning

A compact 2D U-Net [19] F_{coarse} predicts three coarse classes (myocardial ring, LV, RV) from the channel-stacked sequences augmented with constant *presence maps*, i.e. the input is $[\mathbf{I}_{\text{LGE}}, \mathbf{I}_{\text{C0}}, \mathbf{I}_{\text{T2}}, m_1 \mathbf{1}, m_2 \mathbf{1}, m_3 \mathbf{1}]$. Presence maps distinguish missing sequences from dark tissue. Thresholding and largest-component selection give $\hat{Y}^c$ and $P = \mathbf{1}[\hat{Y}^c > 0]$. The dilated bounding box of P defines Stage 2’s region of interest Ω . Cine uses myocardium and LV coarse classes.

4.2 Stage 2a: Multi-Sequence Scar and Anatomy Decoding

Stage 2a predicts scar from LGE-centered features with auxiliary myocardium decoding; Stage 2b handles edema.

Modality-decoupled encoding with presence gating. The framework runs three independent encoders $E^{(m)}$, $m \in \{\text{LGE}, \text{C0}, \text{T2}\}$, each producing a four-level pyramid $f_\ell^{(m)}$, $\ell = 1, \dots, 4$. Missing sequences are suppressed multiplicatively,

$$\tilde{f}_\ell^{(m)} = m_m \cdot f_\ell^{(m)}, \quad (1)$$

so an absent auxiliary sequence contributes no image-derived feature maps; the decoder still receives fixed sigmoid gate values from the zeroed features. The same weights serve LGE-only, LGE+C0 and full three-sequence cases.

An optional alignment branch. Separate breath-holds can introduce fine non-rigid offsets around the thin myocardial ring where lesion decisions are made. The framework therefore includes a deformable branch that models sequence alignment inside the representation the decoder consumes: for each moving sequence $m \in \{\text{C0}, \text{T2}\}$ a localization head $h^{(m)}$ reads the *paired* bottlenecks and

predicts $K = g^2$ control-point offsets $\theta^{(m)} = h^{(m)}([\tilde{f}_4^{(m)}, f_4^{(\text{LGE})}])$, defining a thin-plate spline [2] $\mathcal{T}_\theta$ that warps every pyramid level of that sequence (Appendix A). The final head layer is initialized as described in Appendix A. TPS, prior gating and consistency are enabled in the final scar configuration.

Fusion and prior gating. The decoder fuses the sequence pyramids with a gating rule that keeps LGE as the structural carrier while the other sequences modulate it, and the Stage-1 prior re-enters at every decoder level as a multiplicative gate that emphasizes the cardiac region while retaining context. Appendix B gives the equations and the fixed gate values induced by absent auxiliary sequences.

4.3 A Supervision-Masked Objective for Ragged Annotation

Training pools cases with different annotated class subsets by applying the per-class supervision mask $\mathbf{s}$ directly in the loss normalizer. With $\ell_{b,c}^{\text{dice}}$ and $\ell_{b,c}^{\text{bce}}$ the per-sample per-class Dice and binary cross-entropy terms and w_c a class weight,

$$\mathcal{L}_{\text{seg}} = \frac{\sum_{b,c} s_{b,c} w_c [\ell_{b,c}^{\text{dice}} + \ell_{b,c}^{\text{bce}}]}{\sum_{b,c} s_{b,c}}. \quad (2)$$

Each annotated class contributes its supervised gradient; an unannotated class receives zero weight in both the numerator and the denominator. A case from a scar-focused center therefore supervises the scar head and leaves edema supervision to centers that annotated edema.

Small lesions additionally receive a recall-oriented Tversky term [20] with $\alpha < \beta$ and a focal term [13], both restricted to lesion channels and both masked the same way. Deep supervision is applied at two decoder levels.

An auxiliary *cross-sequence myocardium consistency* term completes Stage 2a (Appendix C). It decodes a myocardium mask from each sequence-specific feature stack and regularizes all available pairs with Dice agreement. This ties representations through shared anatomy without adding annotations.

4.4 Stage 2b: A Dedicated Edema Expert

The dedicated edema expert uses T2-driven evidence with scar and edema supervision available on 220 and 80 cases, respectively.

The framework therefore trains a separate expert. T2 forms the main stream while the LGE and C0 streams are fused element-wise into a *reference* pyramid e_ℓ^{ref} , which enters the decoder *detached*, $d_\ell = \Psi_\ell([\text{up}(d_{\ell+1}), \text{sg}(e_\ell^{\text{ref}}), e_\ell^{\text{T2}}])$. The detached reference supplies anatomical context without receiving gradients through this decoder connection.

The expert regresses the injury zone $Z = \text{edema} \cup \text{scar}$, matching the label representation used for fusion. Pure edema is recovered at fusion time by assigning the predicted scar first and then filling $Z \setminus \widehat{\text{scar}}$ as edema.

4.5 Stage 2c: Motion-Derived Scar from Cine

For cine-only data, the framework replaces the multi-sequence encoder stack with a motion extractor. A shared encoder embeds each of T ED-anchored frames; an anatomy decoder predicts per-frame myocardium and LV; and a motion decoder predicts a dense displacement field between the ED reference I_0 and frame I_t , bounded to keep it physically plausible:

$$\varphi_t = \tau \tanh(D_{\text{mot}}(f_0, f_t)), \quad \varphi_0 = \mathbf{0}, \quad (3)$$

with τ the maximum displacement as a fraction of the field of view. The field warps both the anatomy probability and the frame, $\hat{A}_t = \mathcal{W}(\sigma(A_t), \varphi_t)$ and $\hat{I}_t = \mathcal{W}(I_t, \varphi_t)$.

The pathology head receives only motion, warped anatomy and the anatomical prior:

$$z_t = \Theta\left(\left[\varphi_t, \hat{A}_t, P\right]\right), \quad \hat{Y} = \text{Conv}_{1 \times 1}\left(\frac{1}{T} \sum_t z_t\right). \quad (4)$$

This restricts the cine scar branch to contrast-free temporal and anatomical cues.

Training combines the masked pathology loss with four auxiliary terms: ED-frame anatomy supervision; consistency between each warped anatomy map and the ED prediction; a photometric term matching $\hat{I}_t$ to I_0 ; and an ℓ_1 smoothness penalty on $\nabla \varphi$ for the displacement field.

4.6 Inference

Both regimes share one skeleton: predict anatomy, restrict to the prior, predict pathology with flip test-time augmentation, then apply anatomical containment and minimum lesion-size cleanup. For the multi-sequence regime, scar comes from the Stage-2a scar/anatomy expert, and edema is assigned from $Z \setminus \text{scar}$ under the injury-zone fusion convention. For cine, predictions are additionally averaged over three through-plane resamplings as the deployment-time handling of through-plane resolution variation.

5 Experiments

Data and data-use declaration. All experiments use the CARE-2026 CARE-Myocardium dataset, comprising the MyoPS multi-sequence cohort and the CineMyoPS cine cohort. We use no other data. We gratefully acknowledge the challenge organizers, and cite the sources as required: [28,18,4,3]. Table 1 summarises the heterogeneity that motivates our design. No pre-trained weights are used; every model is trained from scratch.

Implementation. Volumes are resampled to $1.25 \times 1.25 \times 10.0$ mm (multi-sequence) and $1.25 \times 1.25 \times 8.0$ mm (cine), robustly z-scored per volume, and processed slice-wise at 192×192 . Stage-1 uses 24 base channels, Stage-2a 24, the edema expert 64 and the cine model 16, with instance normalization [23].

Table 1. The CARE-2026 CARE-Myocardium training cohort. Sequence availability and annotated classes both vary by center: 116 of 220 multi-sequence cases provide LGE alone, and 80 carry edema labels. The evaluation cohort provides all three sequences, so a model can learn from partial training protocols and exploit complete inputs at test time.

Centre	Cases	Sequences	Annotated classes
<i>MyoPS (multi-sequence)</i>			
A	81	LGE	myo, LV, scar
B	35	LGE+C0+T2	myo, LV, RV, edema, scar
C	45	LGE+C0+T2	myo, LV, RV, edema, scar
E	7	LGE+C0	myo, LV, RV, scar
F	9	LGE+C0	myo, LV, RV, scar
G	8	LGE+C0	myo, LV, RV, scar
H	35	LGE	scar
<i>CineMyoPS (cine only)</i>			
α	40	Cine	myo, LV, scar
β	24	Cine	myo, LV, scar

We train with AdamW [15] under a cosine schedule [14], gradient clipping at 0.1, and a class-quota batch sampler that guarantees a fixed fraction of lesion-bearing slices per batch. The TPS warm-up is $E_w = 10$ epochs. The retained evaluation uses five folds; MyoPS has 176 training and 44 validation cases per fold, and CineMyoPS validation sizes are 13, 13, 13, 13 and 12. Because edema is annotated at two centers, edema metrics are computed on those centers’ cases. Seed 3407 is a configuration default; the exact seeds and training manifests for the final checkpoints remain unresolved.

Metrics. Internal five-fold results are reported as Dice and HD95 (mm), with mean $\pm$ standard deviation across folds. Official validation results are organizer-reported Dice and HD (mm).

5.1 Main Results

Official validation leaderboard. Table 3 reports the best MoSAIC organizer-scored CARE-2026 validation entries for each official target, using the organizer metrics Dice and HD. The best MoSAIC entries reach 0.6965 Dice for MyoPS scar, 0.5983 for MyoPS edema and 0.2069 for CineMyoPS scar. Compared with the nnU-Net comparator, MoSAIC improves MyoPS scar, CineMyoPS scar, and the mean Dice over the three official targets (0.492 to 0.501).

Table 2 retains the author-approved attachment results without additional performance claims. The source of the generated table remains unresolved. These local values are distinct from official validation.

Table 2. Author-retained five-fold results, mean $\pm$ s.d. across folds; table-generation provenance remains unresolved. Pure-edema metrics use cases with edema annotations; the remaining metrics use their corresponding annotated cohorts. Cine scar uses motion and anatomy, without contrast-enhanced input.

Structure	Dice $\uparrow$	HD95 (mm) $\downarrow$
<i>MyoPS: multi-sequence (LGE, C0, T2)</i>		
Myocardium	0.687 ± 0.036	8.29 ± 1.61
LV	0.866 ± 0.015	5.87 ± 1.43
RV	0.841 ± 0.022	6.63 ± 1.80
Pure edema	0.349 ± 0.065	20.21 ± 3.49
Scar	0.668 ± 0.022	11.69 ± 1.06
Lesion union	0.645 ± 0.024	15.87 ± 3.13
<i>CineMyoPS: cine only</i>		
Myocardium	0.695 ± 0.024	5.35 ± 0.30
LV	0.919 ± 0.007	3.87 ± 0.55
Scar	0.457 ± 0.038	19.68 ± 3.88

Table 3. Best organizer-scored MoSAIC entries and the nnU-Net comparator on the CARE-2026 validation leaderboards, reported as Dice/HD (mm). For each target, the MoSAIC row is the submitted entry with the highest Dice.

Leaderboard	Target	MoSAIC Dice/HD	nnU-Net Dice/HD
MyoPS	scar	0.6965 / 13.7827	0.6258 / 14.9844
MyoPS	edema	0.5983 / 26.7067	0.6691 / 21.0898
CineMyoPS	scar	0.2069 / 48.7463	0.1816 / 52.9706

5.2 Discussion

MoSAIC combines anatomy conditioning, input-presence gates and case-aware supervision. Its best official entries improve MyoPS and CineMyoPS scar Dice relative to nnU-Net, with three-track mean Dice increasing from 0.492 to 0.501. The per-track best-Dice entries do not establish mean HD improvement.

Limitations. The retained five-fold table lacks recovered table-generation provenance. Full edema-stage folds 1–4 and local CineMyoPS prediction/ground-truth evaluation artifacts remain unavailable in the evidence package. Clean ablations and controlled sequence-withdrawal evaluation are also missing; module-causal gains, controlled modality robustness, state-of-the-art performance and cross-center robustness are not established. Cine folds are drawn from the same source centers, whereas official validation may change scanner, frame count, temporal resolution and slice thickness; protocol-normalized flow or explicit frame-rate augmentation is therefore a natural extension. For edema, the current fusion predicts $Z = \text{edema} \cup \text{scar}$ and obtains pure edema by subtracting scar at fusion time. Further work will focus on strengthening edema supervision, calibrating edema-zone fusion and adding inter-slice context.

6 Conclusion

MoSAIC uses coarse anatomy to condition pathology decoding, with input presence gates, supervision masking, an edema-zone expert and cine motion evidence. The best official validation Dice scores are 0.6965 for MyoPS scar, 0.5983 for MyoPS edema and 0.2069 for CineMyoPS scar. These results support the reported validation endpoints without resolving the limitations of the retained internal evaluation.

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A Thin-Plate Spline Alignment Field

With target control points $\{p_k\}$ on a regular $g \times g$ lattice spanning $[-r, r]^2$ and predicted offsets $\theta_k^{(m)}$, the thin-plate spline [2] is the interpolant of $\{p_k \mapsto \theta_k^{(m)}\}$

with minimal bending energy:

$$\mathcal{T}_\theta(q) = a_0 + Aq + \sum_{k=1}^K w_k U(\|q - p_k\|), \quad U(d) = d^2 \log d, \quad (5)$$

where (w, a_0, A) solve the standard TPS linear system. That system depends only on $\{p_k\}$, which are fixed, so it is inverted once and cached. The final linear layer of the localization head $h^{(m)}$ is zero-initialized with its bias set to the target lattice, giving a stable starting warp that is updated by the task loss.

B Cross-Modality Gated Fusion

The Stage-2a decoder fuses the aligned pyramids at level ℓ as

$$a_\ell = \frac{1}{2} \left[\sigma(\hat{f}_\ell^{(C0)}) + \sigma(\hat{f}_\ell^{(T2)}) \right], \quad (6)$$

$$d_\ell = \Phi_\ell \left(\left[\text{up}(d_{\ell+1}), \quad f_\ell^{(\text{LGE})} \odot a_\ell, \quad \sigma(\hat{f}_\ell^{(C0)}), \quad \sigma(\hat{f}_\ell^{(T2)}) \right] \right), \quad (7)$$

initialized from the concatenated bottlenecks $d_4 = [f_4^{(\text{LGE})}, \hat{f}_4^{(C0)}, \hat{f}_4^{(T2)}]$. When an auxiliary sequence is absent, its masked feature pyramid is zero, so its sigmoid gate channel is fixed at $\sigma(0) = \frac{1}{2}$. This is a consequence of the gate parameterization, not an identity gate; the absent sequence contributes no image-derived feature map. The Stage-1 prior gates every decoder level, $d_\ell \leftarrow d_\ell \odot \sigma(\text{pool}(P))$.

C Auxiliary Training Objectives

Cross-sequence myocardium consistency. Three auxiliary heads $g^{(m)}$ decode a myocardium mask from each sequence’s own features. For the available sequences $A = \{m : m_m = 1\}$, each head is supervised against the myocardium label and every available pair in $\mathcal{Q} = \{(m, m') : m < m', m, m' \in A\}$ is driven into agreement:

$$\mathcal{L}_{\text{cons}} = \frac{\mathcal{L}_{\text{sup}} + \mathcal{L}_{\text{pair}}}{N}, \quad N = |A| + |\mathcal{Q}|. \quad (8)$$

Here $\mathcal{L}_{\text{sup}}$ sums the sequence-specific myocardium Dice terms, and $\mathcal{L}_{\text{pair}}$ sums the pairwise Dice agreement terms.

Stage-2b edema expert. The reference pyramid is formed as

$$e_\ell^{\text{ref}} = \max(E_\ell^{\text{LGE}}([\mathbf{I}_{\text{LGE}}, M]), E_\ell^{\text{C0}}([\mathbf{I}_{\text{C0}}, M])), \quad e_\ell^{\text{T2}} = E_\ell^{\text{T2}}([\mathbf{I}_{\text{T2}}, M]), \quad (9)$$

with M a dilated cardiac mask from Stage 1.